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Observational evidence of elevated smoke layers during crop residue burning season over Delhi: Potential implications on associated heterogeneous $PM_{2.5}$ enhancements

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ABSTRACT

Post monsoonal agricultural Crop Residue Burning (CRB) over northwestern India is believed to severely affect the air quality of the megacity of Delhi. However, the mechanistic understanding remains elusive. Long-term satellite observations (2007–2020) of aerosol properties during CRB season (Oct 20th to Nov 20th) indicate a distinct airshed of CRB plume transport from Northwestern India (source region) to greater Delhi (downwind region). Theoretically, the smoke concentration should disperse downwind, and the CRB-associated $PM_{2.5}$ enhancement over Northern India should be inversely proportional to the distance from the CRB source region. However, the in-situ $PM_{2.5}$ observations illustrate that smoke-associated enhancement in $PM_{2.5}$ over greater Delhi (downwind region) is disproportionately large compared to the source region. In this study, we examine satellite, radiosonde, and ground-based observations along with reanalysis data to provide robust evidence that aerosol- boundary layer - $PM_{2.5}$ associations via semi-direct effect can explain the above heterogeneity. Vertically resolved satellite observations indicate that as the emitted CRB-smoke plumes travel downwind, a portion of the transported smoke plume injected relatively at a higher altitude leads to the formation of an elevated smoke layer over greater Delhi. These elevated smoke layers tend to suppress the mixing height via inducing strong temperature inversion, thereby severely constraining the daytime dilution effect of shallow boundary layers. Along with direct advection of smoke particles, enhanced accumulation of local urban emissions due to these semi-direct impacts can lead to the observed disproportionate $PM_{2.5}$ enhancements over Delhi during CRB haze periods. Thus, control of the local anthropogenic emissions could bring relief during the extreme haze episodes over Delhi.

1. Introduction

Extremely severe haze episodes associated with the agricultural crop residue burning (CRB) is witnessed over northern India during mid-October to mid-November, hereafter referred to as the CRB season (Singh and Kaskaoutis, 2014; Jethva et al., 2018a; Sarkar et al., 2018; Chowdhury et al., 2019; Jethva et al., 2019; Shyamsundar et al., 2019; Mhawish et al., 2020). The farmers in northwestern India and subcontinent burn the residue of the crops for cleaning and preparing the fields for the next harvest, and it has been a common practice in this region for several decades (Shaik et al., 2019; Bilal et al., 2021; Liu et al., 2021; Mhawish et al., 2021). Climatologically, the spatial hotspot of these CRB activities in post-monsoon is centered over Northwestern India, including states of Punjab and Haryana. In the last decade, particularly after implementing the groundwater preservation policy in 2009 (Punjab Sub-oil Preservation Act, 2009) (Singh, 2009), the seasonal shift in CRB in the northwestern India region led to an increase in fire intensity to later post-monsoon compared to that prior to 2009 (Jethva et al., 2019; Sembhi et al., 2020; Liu et al., 2018). Massive emissions of particulate matter (PM) from CRB fires, shallower planetary boundary layer

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height (PBLH), and weaker winds during early winter are often linked to these frequent haze/fog formations over Northern India (Apte et al., 2015; Jethva et al., 2018a; Mukherjee et al., 2020; Sembhi et al., 2020; Mhawish et al., 2021). The prevailing air circulation in the boundary layer is predominantly north-westerly during the CRB season (Kaskaoutis et al., 2014; Vijayakumar et al., 2016), which advects the emitted CRB smoke from its source region downwind towards the heavily populated and urbanized eastern Indo Gangetic Plains (IGP) region (Kaskaoutis et al., 2014; Vijayakumar et al., 2016; Takigawa et al., 2020). Consequently, poorer air quality and adverse health impacts over the Indian megacity of Delhi is associated with these CRB episodes (Badarinath et al., 2009; Anu Rani Sharma et al., 2010; Kaskaoutis et al., 2014; Sahu et al., 2015; Vijayakumar et al., 2016; Cusworth et al., 2018; Chowdhury et al., 2019; Kanawade et al., 2019; Ningombam et al., 2019; Shaik et al., 2019; Kulkarni et al., 2020; Mhawish et al., 2020; Ojha et al., 2020; Takigawa et al., 2020; Patel et al., 2021). Prolonged exposure to fine PM affects human health, leading to cardiovascular and respiratory diseases (Ghude et al., 2016; Lelieveld et al., 2018). Thus, a mechanistic understanding of these extreme pollution episodes is warranted for any mitigation and management strategy, especially over Delhi.

Recent studies have investigated the spatial distribution of aerosol optical depth (AOD) and associated chemical, optical and microphysical properties (Vadrevu et al., 2011; Pant et al., 2015; Kumar et al., 2018; Sarkar et al., 2018; Sawlani et al., 2018; Shaik et al., 2019; Vinjamuri et al., 2020; Mhawish et al., 2021). However, due to the lack of adequate in-situ measurements over the region, the spatial distribution of CRB-associated changes in $PM_{2.5}$ is still not clear. Largely, the $PM_{2.5}$ (particulate matter with an aerodynamic diameter $<$2.5 μm) concentration sees 2–3 folds enhancement (to ~200–250 $\mu g m^{-3}$) compared to the pre-burning period (~80–150 $\mu g m^{-3}$; (Satish et al., 2020)) and pre-monsoon (MAM) wheat burning season (~20–125 $\mu g m^{-3}$;(Rajput et al., 2014)) over the CRB source region (Punjab state). (Rastogi et al., 2014) reported that the emission from biomass burning contributed around 64% of the total $PM_{2.5}$ concentration during the CRB period in Patiala during a campaign study in 2011. In contrast, the daily mean $PM_{2.5}$ concentrations over greater Delhi, a downwind receptor region, get enhanced by 4–6 times and even reaches up to ~500–600 $\mu g m^{-3}$ during this period compared to its background values (~50–100 $\mu g m^{-3}$) (Jethva et al., 2018a; Chowdhury et al., 2019; Mhawish et al., 2020). Another recent study has illustrated that CRB-associated enhancement of $PM_{2.5}$ over Delhi regions is also greater than the other upwind CRB source region of the Haryana state (Saxena et al., 2021). In Delhi, a recent study also showed that the PM_1 (PM with an aerodynamic diameter $<$1 μm) could exceed 1000 $\mu g m^{-3}$ during peak hours with a significant contribution from organic carbon, mainly from CRB (Patel et al., 2021). Recently, (Lalchandani et al., 2021) estimated ~ (44%–53%) of organic aerosols measured in Delhi was influenced by long-range transported biomass burning emissions during post-monsoon season. Note that few of these reported $PM_{2.5}$ values have either been case studies or field campaigns with limited spatial coverage or spanning a few weeks. Nonetheless, the evidence of CRB-associated enhancement in $PM_{2.5}$ concentration over the downwind region (Delhi) being greater than the same over the source region (Punjab-Haryana) is counter-intuitive and cannot be explained by only smoke advection/dispersion mechanisms. Using back trajectory analysis (Jethva et al., 2018a) suggested that self-lofting smoke plume over the CRB region might explain the lower $PM_{2.5}$ enhancements over the source areas. So, are these smoke plumes descending as they travel downwind towards Delhi. Thus, our process-level understanding of these pollution events is limited.

The reported huge enhancement in $PM_{2.5}$ over Delhi ($>$250 km downwind from the source region) can be hypothesized as strong coupling and feedbacks between the dispersing smoke plume and local meteorology. Light-absorbing carbonaceous aerosols like smoke can suppress the evolution of planetary boundary layer height (PBLH) via two simultaneous pathways and consequently affect the local air quality under stagnant weather (Ding et al., 2013; Li et al., 2017; Zheng et al., 2017; Huang et al., 2018). Firstly, these smoke particles can absorb solar radiation and induce a strong dimming effect at the surface, thereby weakening the surface heat fluxes (Ding et al., 2013; Quan et al., 2013; Guo et al., 2016; Li et al., 2017). Secondly, these absorbing aerosols in the atmosphere could heat the atmosphere by trapping incoming solar energy and results in vertical temperature stratification, hence less turbulence (Ding et al., 2016, 2017). The processes, as mentioned above, also known as the 'Dome effect', are said to enhance substantially the near-surface haze pollution (Ding et al., 2016, 2019; Wang et al., 2018). Using simulations, a few recent case studies have illustrated that absorbing aerosols can affect the PBLH over IGP and hence the $PM_{2.5}$ enhancement during the winter season and CRB events (Bharali et al., 2019), but observational evidences during CRB episodes are still rare. Nonetheless, the real question is whether these aerosol-PBLH feedbacks explain the greater enhancement of $PM_{2.5}$ over the downwind region compared to the CRB source region. A skeptical argument against this possibility is that the strength of the smoke-PBLH association associated with CRB emissions should be greater over the source region (smoke plumes will be concentrated) compared to the receptor region.

Here, we used long-term data from multi-satellite observations (2007–2020), ground-based in-situ stations (2015–2020), India Meteorological Department (IMD) radiosonde profiles (2007–2020), and reanalysis product to systematically address the following objectives: 1) Detailed analysis of the spatiotemporal pattern in CRB-associated enhancement in surface $PM_{2.5}$ concentration over northwestern India, 2) Investigate the differences in CRB smoke plume characteristics between the source and downwind receptor region 3) examine the role of smoke-meteorology-$PM_{2.5}$ coupling in explaining the disproportionate enhancement of CRB-associated in surface $PM_{2.5}$ concentration over downwind region, specifically Delhi.

2. Methods and datasets:

2.1. Ground-based $PM_{2.5}$ measurements

The hourly averaged ground-level $PM_{2.5}$ concentrations were obtained from 42 air quality monitoring stations distributed across the Indian state of Punjab, Haryana, Delhi, and western Uttar Pradesh from 2015 to 2020. The $PM_{2.5}$ data were obtained from 41 stations operated by the Central Pollution Control Board (https://app.cpcbccr.com/ccr/#/caaqm-dashboardall/caaqm-landing) from 2018 to 2020 and additional operated by the US embassy in New Delhi from 2015 to 2020 (https://www.airnow.gov/international/us-embassies-and-consulates). Only stations with daily mean $PM_{2.5}$ $>$ 28 measurements per year (90%) from 20th Oct-20th Nov were used in the analysis. The $PM_{2.5}$ is measured with beta gauge attenuation monitors (BAM-1020; Met One Instruments) and reports hourly mean $PM_{2.5}$ concentrations (Mhawish et al., 2020).

2.2. Satellite-based measurements

Several satellite-based aerosol datasets from passive and active sensors, including Moderate Resolution Imaging Spectroradiometer (MODIS), Ozone Monitoring Instrument (OMI), and Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO), were used in this study. Following are the dataset obtained from each satellite sensor:

2.2.1. MODIS dataset

In the present study, Aqua and Terra MODIS collection 6 Multi-Angle Implementation of Atmospheric Correction (MAIAC) AOD data product (MCD19A2 C6) (Lyapustin et al., 2018) and active fire count data (MCD14DL) (Giglio et al., 2016) from 2007 to 2020 are obtained from LAADS DAAC (https://ladsweb.modaps.eosdis.nasa.gov/) and NASA's Fire Information for Resource Management System (FIRMS) (https://firms.modaps.eosdis.nasa.gov). MAIAC aerosol retrieval algorithm retrieves AOD at 1 km spatial resolution over all land surfaces except snow/ice and ocean (Lyapustin et al., 2018). Our previous study over South Asia (Mhawish et al., 2019) showed that the MAIAC AOD at 550 nm outperform other MODIS aerosol retrieval algorithms Dark Target (Levy et al., 2013) and Deep Blue (Hsu et al., 2013) in terms of retrieval accuracy and the ability to retrieve AOD in small scale in between the cloud and AOD for thick smoke layers. Similar good agreement between MAIAC AOD_{470} vs AERONET AOD_{470} was also reported over Kanpur city in the middle of the IGP region (Jethva et al., 2019). The highest quality flag assurance (QA) cloud mask value "clear" MAIAC L2 AOD_{550} data and the Aqua and Terra MODIS active fire count with confidence >80% were considered in the analysis.

2.2.2. OMI measurements

Ozone monitoring instrument (OMI) is a hyperspectral imaging spectrometer onboard of Aura satellite and measures the backscatter solar radiation in UV to visible spectrum range (270–500 nm). The multiple UV to visible bands allow for retrieving absorbing aerosol types such as carbonaceous aerosols and mineral dust (Torres et al., 2007, 2013). Aerosol absorption optical depth (AAOD) at 388 nm and UV-Aerosol Index (UV-AI) are key aerosol parameters retrieved by OMI OMAERU aerosol retrieval algorithms and are widely used to quantify and identify the absorbing aerosols in cloud-free conditions (OMAERU) (Torres et al., 2018) as well as cloudy scenes (OMACA) (Jethva et al., 2018b). In the current analysis, OMAERUV Level 2 (L2G) version 3 AAOD and UV-AI products binned on to each 0.25° x 0.25° were used. We have considered the AAOD and UV-AI retrievals with quality flag 0 and 1 when AAOD retrieval is reliable (Jethva and Torres, 2011).

2.2.3. CALIOP measurements

Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP) is two-wavelength (532 nm, and 1064 nm) satellite-based LiDAR onboard Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO) satellite launched in 2006 to study the vertical structure of atmospheric aerosols and clouds. CALIOP measures the vertical profile of attenuated backscatter radiation with a vertical resolution of 30 m (from surface to 8.2 km) and 60 m (from 8.2 to 20 km). The CALIOP aerosol retrieval algorithm retrieves extinction coefficient at 5 km spatial resolution and 30–60 m vertical resolution. The algorithm has continuously been upgraded to reduce the uncertainties and improve the sensitivities of aerosol detection (Kim et al., 2018). In the current analysis, we used the latest Version 4.2 (V4.2) Level 2 (L2) of the vertical profile of aerosol extinction coefficient at 532 nm and 1064 nm from 2007 to 2020. Detailed information about the CALIOP aerosol retrieval algorithm can be found in (Kim et al., 2018).

2.3. Radiosonde measurements

Daily radiosonde observations from 2007 to 2020 were obtained from the land-based radiosonde station network of the University of Wyoming database archive (Station ID: 42182 VIDD/New Delhi; http://weather.uwyo.edu/upperair/sounding.html). The radiosonde measures the temperature profile twice a day (00:00 UTC and 12:00 UTC). In this analysis, the virtual potential temperature profile measurements at 12:00 UTC (17:30 IST) were used to estimate the mixing layer height and changes in the vertical profile of the atmospheric temperature. The radiosonde-based mixing layer height over Delhi was calculated using the gradient of virtual potential temperature (θv) method 'Parcel Method' (Holzworth, 1964; Seibert, 2000; Seibert et al., 2002). The vertical differences in the potential temperatures are relatively symmetric within the mixing layer, and the values are close to the near-surface potential temperature. The significant gradient change in the potential temperature considers the boundary of the mixing height.

2.4. MERRA-2 reanalysis

Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) is the latest NASA's atmospheric reanalysis data released in 2017 by Global Modeling and Assimilation Office (GMAO) (Buchard et al., 2017; Randles et al., 2017). MERRA employs upgraded Goddard Earth Observation System, version 5 (GEOS-5) coupled to the Goddard Chemistry Aerosol Radiation and Transport Model (GOCART) aerosol module and simulates aerosol species including dust, sea salt, sulfate, black carbon, and organic carbon. The spatial resolution of MERRA-2 data is 0.5° x 0.625° and 72 hybrid-eta levels from the surface to 0.01 hPa. This analysis used MERRA-2 aerosol species black carbon, organic carbon, and air temperatures at 940hpa and 985hpa pressure levels. Additionally, the ground-level $PM_{2.5}$, atmospheric boundary layer height, and air temperature tendency due to shortwave profile, surface net downward shortwave flux assuming clear-sky (SWGNTCLR), surface net downward shortwave flux assuming clear-sky and no aerosol (SWGNTCLRCLN), surface net downward longwave flux assuming clear-sky (LWGNTCLR), surface net downward longwave flux assuming clear-sky and no aerosol (LWGNTCLRCLN) are also used in the current analysis.

2.5. Methodology and data processing

The methodology of our analysis can be divided into three steps. First, based on long-term satellite data, including AOD, AAOD, UV-AI, and active fire count, we studied the CRB climatology of fire and aerosol loading over the region. Our analysis focuses on CRB season i.e. October 20th to November 20th, when most crop residue burning occurs (51% -82%) for 2007 to 2020 (Table S1). The spatial distribution of the fire activities showed the highest detected fire in Punjab and Haryana states (Fig. 1a, b). On the other hand, based on the long-term climatology of satellite-based aerosol properties, high AOD_{550}, $AAOD_{388}$, and UV-AI were observed from the source region to western Uttar Pradesh state. Therefore, we described this trajectory as a CRB smoke path (CRB_{box}) that starts from the source with high fire activities to the downwind receptor region (Fig. 1). The smoke trajectory is further divided into two boxes. The first box represents the source region while the second represents the downwind region (Fig. 1 e) to study the variation of aerosols between both source and sink regions.

The second step is to quantify the CRB associated changes on ground-level $PM_{2.5}$, columnar aerosol properties, vertical distribution of aerosol, and atmospheric mixing height by composite analysis method of heavily smoky days against background days. Using daily samples of mean AOD values within the CRB_{box} during the CRB period, a threshold AOD value for smoky days (AOD percentile >75 percentile) and non-smoky or background days (AOD percentile <25 percentile) was determined. Although CRB burning may be happening every day, we used this strict stratification method for segregating our samples into heavy smoky days and less smoke days (i.e background days) to compare and contrast the $PM_{2.5}$ environment of heavy haze days than relatively most clear days. Using these threshold values, the mean $PM_{2.5}$ concentration corresponding to smoky and background days has been calculated separately at all monitoring sites. The ground-based $PM_{2.5}$ measurements were used to understand the spatial distributions and diurnal variability associated with CRB during smoky and background days. The vertically resolved CRB smoke is also examined using satellite-based Lidar retrieval to understand the variability in aerosol vertical profile during smoky and background days over source and downwind regions. CALIOP extinction coefficients retrieval at 532 nm and 1064 nm were examined for smoky and background days over source and downwind regions. Finally, the atmospheric stability was examined using radiosonde measurements in Delhi during smoky and background days. The radiosonde-based virtual potential temperature (VTP) profiles correspond to smoky and background days used to understand the impact of CRB smoke plumes on atmospheric stability. To further understand the

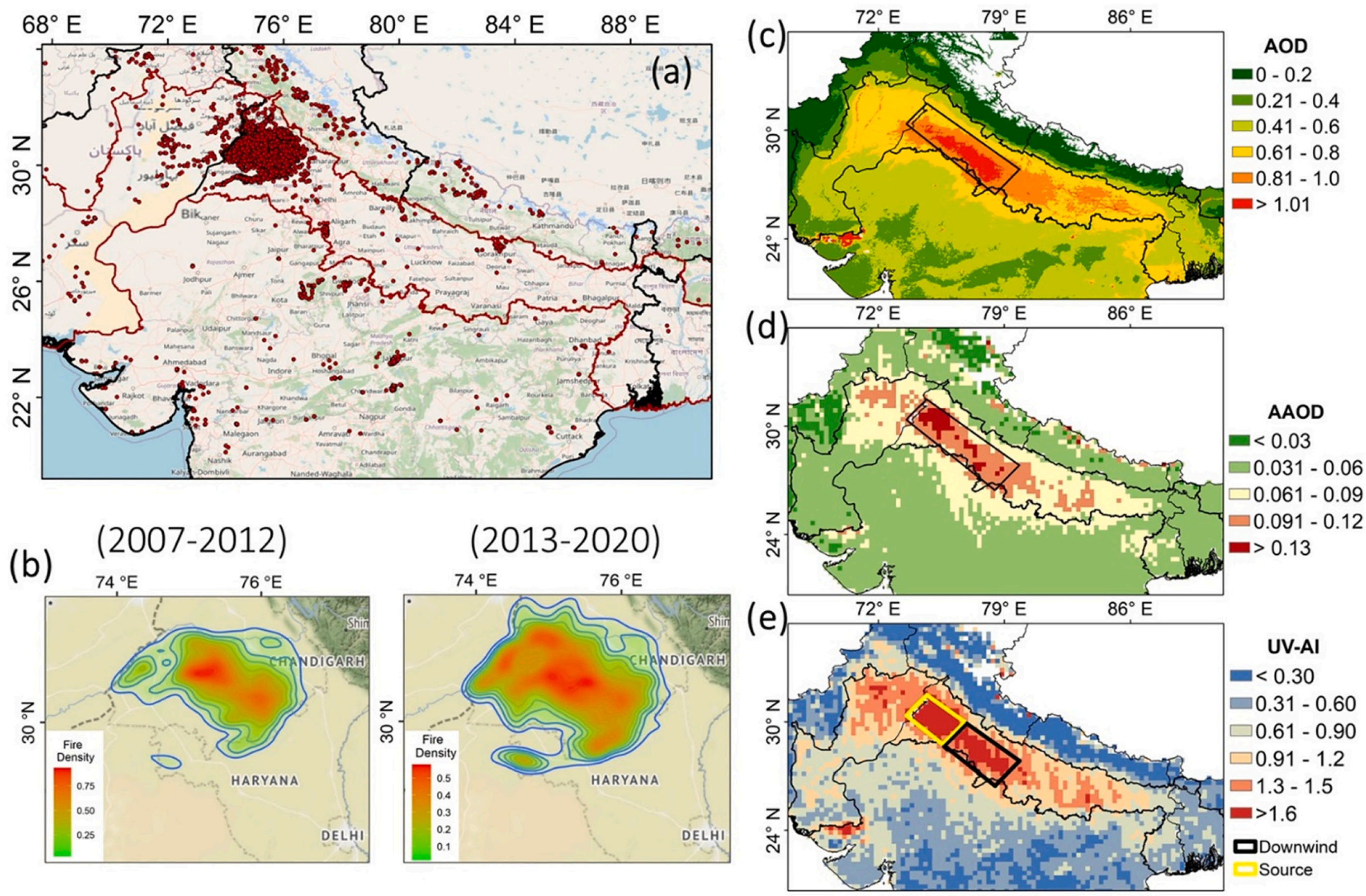

Fig. 1. (a) Spatial distribution of 14 years (2007–2020) mean active fire count during CRB period, (b) heat map of fire density using 2D kernel density estimation, (c) AOD, (d) AAOD, and (e) Aerosol Index (UV-AI) over northern Indian subcontinent during CRB season (October 20th to November 20th) from 2007 to 2020. The red boundary in (a) represents the IGP region and the black box in (c) represents the CRB smoke path (CRB_{Box}), while the two boxes in (e) represent the source region (blue) and the downwind region (brown). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

impact of mixing height on $PM_{2.5}$ concentration, the relationship between collocated radiosonde-based mixing height and ground-based $PM_{2.5}$ was examined over the downwind region (Delhi).

Finally, MERRA-2 simulated data over the Megacity Delhi is used to verify our observation-based hypothesis of elevated aerosol layers induced enhancement in $PM_{2.5}$. MERRA-2 simulated carbonaceous species and air temperatures at 940hpa and 985hpa pressure levels have been used to calculate the ratio of BOC at ~940hpa to the near-surface layer at 985hpa (BOC_{ratio}) and the air temperature differences between ~940hpa and ~ 985hpa (ΔTemp) using Eqs. (1) and (2) respectively.

$$BOC_{ratio} = \frac{(BC+OC)_{940hpa}}{(BC+OC)_{985hpa}} \tag{1}$$

Where, BC and OC represent black carbon and organic carbon mixing ratio (kg/kg), respectively.

$$\Delta Temp = (Temp)_{985hpa} - (Temp)_{940hpa} \tag{2}$$

where, Temp represents the MERRA-2 air temperature (K).

The aerosol direct radiative forcing (ARF) has been calculated using MERRA-2 radiative variables (i.e., SWGNTCLR, SWGNTCLRCLN, LWGNTCLR, and LWGNTCLRCLN). The ARF is defined as the difference between the solar flux in the presence of aerosol and absence of aerosol (AOD = 0). The ARF (W/m^{-2}) was calculated using the following equation (Eq. (3)) and the aerosol radiative forcing efficiency (ARF_{eff}) determined by radiative forcing (W/m^{-2}) per unit AOD (Eq. (4)).

$$ARF = (SWGNTCLR + LWGNTCLR) - (SWGNTCLRCLN + LWGNTCLRCLN) \tag{3}$$

$$ARF_{eff} = \frac{ARF}{AOD} \tag{4}$$

The shortwave heating rate profile was obtained from MERRA-2 air temperature tendency due to shortwave (Ks^{-1}). The shortwave heating rate as a function of altitude can provide information on the heat exchange due to absorbing aerosol types in the atmospheric column (Feng et al., 2016).

The regression analysis used in this study was performed using the Reduced Major Axis (RMA) regression method that incorporates the error in both dependent and independent variables (Smith, 2009). Several studies have used the RMA regression method to estimate the slope of the best fit line between different estimates measured with errors such as data obtained from remote sensing retrieval and model simulation (Li et al., 2018; Bilal et al., 2021).

3. Results

3.1. The CRB emissions and the airshed of Delhi

Climatologically, the northwestern Indian states of Punjab and Haryana witnessed a high frequency of active fire counts during the CRB period, i.e., October 20th to November 20th, clearly behaving as the source region of CRB emissions in India (Fig. 1a & S1). Moreover, the spatial heat maps of CRB fires for two time periods 2013–2020 and

2007–2012 (Fig. 1b-c), show that the fire density has enhanced over the last decade (Jethva et al., 2019; Liu et al., 2021). The spatial variation of mean AOD and aerosol absorption optical depth (AAOD) values during the CRB period shows extremely high columnar aerosol loading (AOD >1.01) from the source region of Punjab, Haryana states to central IGP, via Delhi (Fig. 1c; identified by black bounding box). Further, distinctly high values of AAOD (>0.1) and UV-AI (>1.5) are also present within this region additionally suggests the dominance of absorbing aerosols within the box. Previously, (Liu et al., 2018) have also identified a similar bounded smoke airshed for Delhi using back trajectory analysis and showed that variations in fire and meteorology over the source CRB region can contribute to ~40% variance in PM anomalies over Delhi within 5 days.

The mean AOD, AAOD and UV-AI values within the airshed box are found to be ~40–60% higher in magnitudes in comparison to the adjoining regions i.e., states North to the Himalayan foothill, eastern Uttar Pradesh, Bihar, West Bengal and Pakistan. Although, the variation in columnar aerosol loading (AOD) within the airshed box was not substantial, the magnitude of absorbing aerosol loading (AAOD and UV-AI) over source region was ~14% greater compared to downwind region (Table S2). This strongly underlines the distinction of the box as a strategic region for the contemporary air quality challenges over India. For example, cities in Punjab like Amritsar, Jalandhar, and Chandigarh, located within a 200 km radius of the CRB sources (but outside the CRB_{Box}), have relatively lower AOD values (0.6–0.8) in comparison to the cities located at 300–500 km downwind of the CRB region within the CRB_{Box}. Further, the mean AOD values within this box vary linearly with the total active fire counts (Fig. S2d), and the CRB-associated enhancement in AOD within this box can be as high as 0.7–1.0 (Fig. S2 a-c). Thus, it is reasonable to interpret that most smoke plumes travel from the CRB source region to the airshed of the downwind receptor sites like Delhi via this boxed region, hereafter referred to as CRB_{box}. Our observational interpretation of the CRB smoke path is strongly supported by the simulations performed in (Cusworth et al., 2018; Liu et al., 2018). Using numerous CRB fire emission scenarios, they illustrate that the CRB plumes affecting $PM_{2.5}$ of the Megacity Delhi enter the airshed of Delhi via the same box.

3.2. The CRB associated spatiotemporal changes in surface $PM_{2.5}$

Next, the CRB-induced changes in daily mean surface $PM_{2.5}$ concentrations over northwestern India (specifically the downwind receptor region vs the source region) are investigated (Fig. 2). As expected, the concentrations of $PM_{2.5}$ over all the sites in Northwestern India increased (by 25–200 μgm^{-3}) during smoky days compared to the background (non-smoke) days (Fig. 2a–c). Nonetheless, greater differences in $PM_{2.5}$ concentration (Fig. 2c) were observed (~100–200 μgm^{-3}) for the CRB_{box} sites compared to the sites located outside the box (<25–75 μgm^{-3}). In other words, $PM_{2.5}$ enhancement during smoky days was found to be 100–200% for the sites within the CRB_{box} (relative to the background), but the same is ~50% for the sites located outside the box (Fig. S3a).

The CRB_{box} can be divided into two subregions (Fig. 1e; identified by yellow and black boxes). The first subregion is the source region, covering the area exhibiting a very high frequency of active fire counts, and the second subregion represents the downwind region, containing the parts of Haryana, and the Delhi megacity, characterized by high local anthropogenic aerosol emissions. Although the air quality monitoring stations are unevenly distributed across the region, the spatial heterogeneity in CRB-induced enhancement of $PM_{2.5}$ between source and downwind regions is noteworthy. The satellite's columnar AOD, AAOD, and UV-AI measurements showed higher mean values over the source region than the downwind region during smoky days (Table S2). Theoretically, we would expect a same pattern with a decrease in the

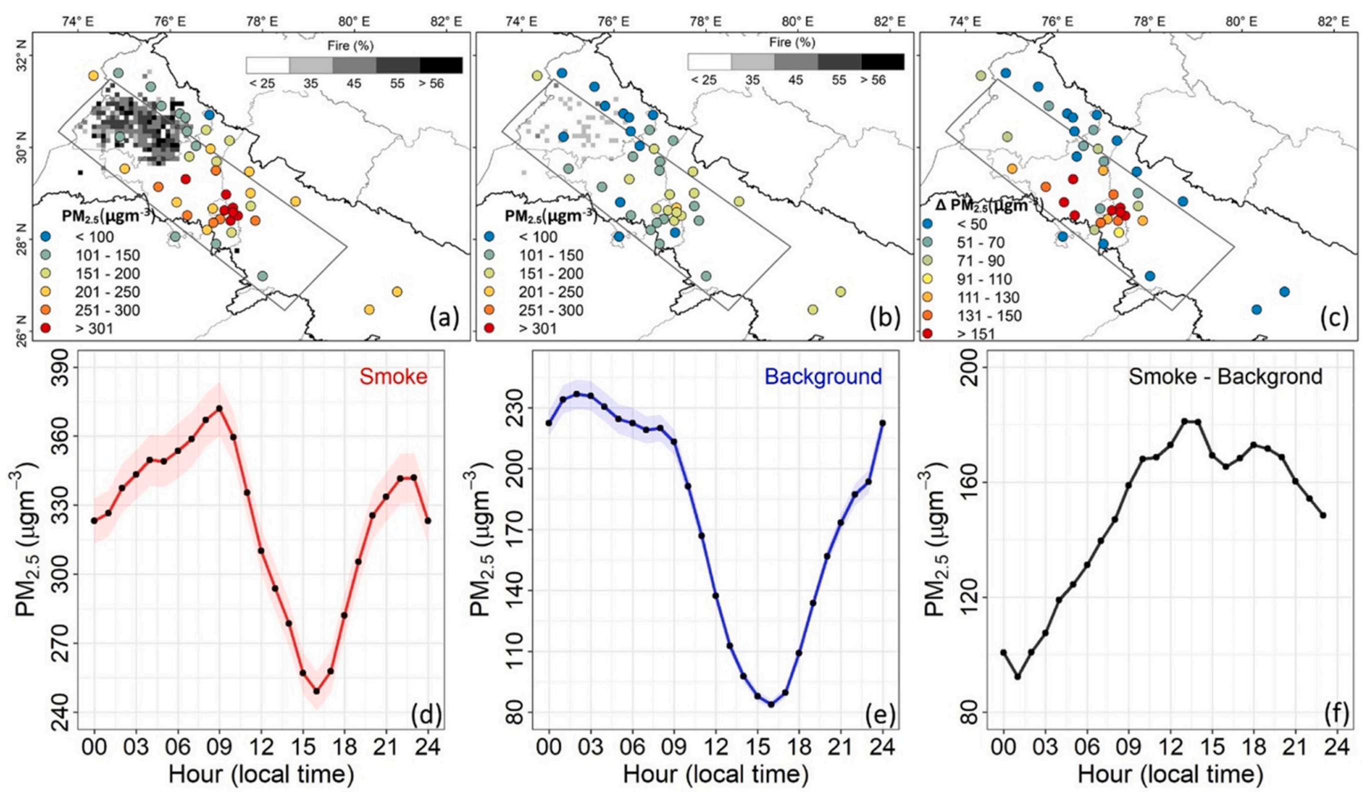

Fig. 2. The spatial distribution of the mass concentrations of $PM_{2.5}$ during a) smoky days, b) background days, c) the $PM_{2.5}$ differences between smoke and background days. The gray background in (2a&b) represents the percentage enhancement in active fire counts during smoky days and background days, respectively. Fig. 2d & e represent the diurnal variation of $PM_{2.5}$ during smoke (d) and background (e) days, and shades in (d&e) represent the 95% confidence interval in the mean $PM_{2.5}$ concentrations, while (2f) represents the diurnal variation of $PM_{2.5}$ difference between smoke and background days.

impact of CRB emissions on surface $PM_{2.5}$ as we move farther from the source region. In agreement, a slight decrease in the magnitude of percentage change in the $PM_{2.5}$ with distance from the CRB hotspot is observed within the CRB_{box} (Fig. S3a). However, the absolute value of CRB-associated enhancement in surface $PM_{2.5}$ increases with distance from Punjab to Delhi within the CRB_{box}. While the enhancement is over Punjab (100 km downwind) and Haryana (200 km downwind) sites are ~50–100 and 100–200 μgm^{-3}, respectively (Fig. 2c), the same over the Delhi NCR region (350 km downwind) is ~150–300 μgm^{-3}. This spatial understanding of 3-year-mean values strengthens the previous reports of disproportionate enhancement in $PM_{2.5}$ associated with CRB emissions, as discussed in Section 1.

Further, we examine the diurnal variation of $PM_{2.5}$ (averaged over the sites within CRB_{box}) during smoky (Fig. 2d) and background (Fig. 2e) days. As expected, the hourly $PM_{2.5}$ values are greater during nighttime and decrease as the day progress from morning to evening due to the dilution effect of the boundary layer evolution (see diurnal cycle of PBLH in Fig. S4). Quantitatively, on average, the maximum values (~230 μgm^{-3}) were observed during midnight, and the minimum values (~80 μgm^{-3}) were observed during the late afternoon. As such, a prominent diurnal range was observed in the $PM_{2.5}$ values (max $PM_{2.5}$ - min $PM_{2.5}$) during background days (~ 152 μgm^{-3}). In contrast, the hourly $PM_{2.5}$ distribution for smoky days illustrates that peak $PM_{2.5}$ values are found during daytime (rather than nighttime). Specifically, the $PM_{2.5}$ values increased from midnight and reached the peak at 0900 LT (~372 μgm^{-3}), beyond which the values started to decrease. The minimum value is still observed during the late afternoon, but the absolute magnitude is very high (~240 μgm^{-3}). For more clarity, we illustrate the hourly differences in mean $PM_{2.5}$ values between smoky and non-smoky days (Fig. 2F). It highlights that the CRB-associated enhancement of $PM_{2.5}$ over the downwind region mainly occurs during daytime, with peak enhancement (~180 μgm^{-3}) occurring around Noon. Consequently, the mean diurnal range is comparatively lower during smoky days (mean value ~122 μgm^{-3}).

The change in diurnal range is even more prominent when we regress the daytime $PM_{2.5}$ values (1200–1800) with corresponding nighttime values (0300–0900). As the daytime $PM_{2.5}$ values distribution enhances from 50 to 150 μgm^{-3} during background days to 100–300 μgm^{-3} during smoky conditions, the slope of RMA regression during smoky days is half of that during background days (Fig. S3b), indicating very high public exposure time to hazardous levels of $PM_{2.5}$ throughout the entire day.

3.3. *The vertical distribution of smoke over the source and downwind regions*

This section describes the vertical distribution of aerosols using observed aerosol extinction coefficient (*E*xt. Coef.) profiles at 532 nm and 1064 nm from CALIPSO satellite overpasses during the CRB period. Similar to the classification used in Fig. 2d-f, the vertical profiles are segregated into smoky days and background days (discussed in the above section). We assume that the aerosol extinction coefficient of smoky days is a proxy for CRB smoke plumes within CRB_{box}. During background days, the near-surface (below 200 m) mean extinction coefficient values at 532 nm and (1064 nm) are ~1.0 km^{-1} and ~ 1.5 km^{-1} (~0.23 km^{-1} and ~ 0.74 km^{-1}) over the source and downwind region, respectively (Fig. 3 a&b and Fig. S5). The relatively higher near-surface extinction coefficient observed at the downwind region is due to the greater surface emission rates from anthropogenic activities over the greater Delhi region. This is consistent with the spatial pattern observed for $PM_{2.5}$ concentrations and columnar AOD during the background days (See Fig. 2b and Table S2). Also, irrespective of the region, the

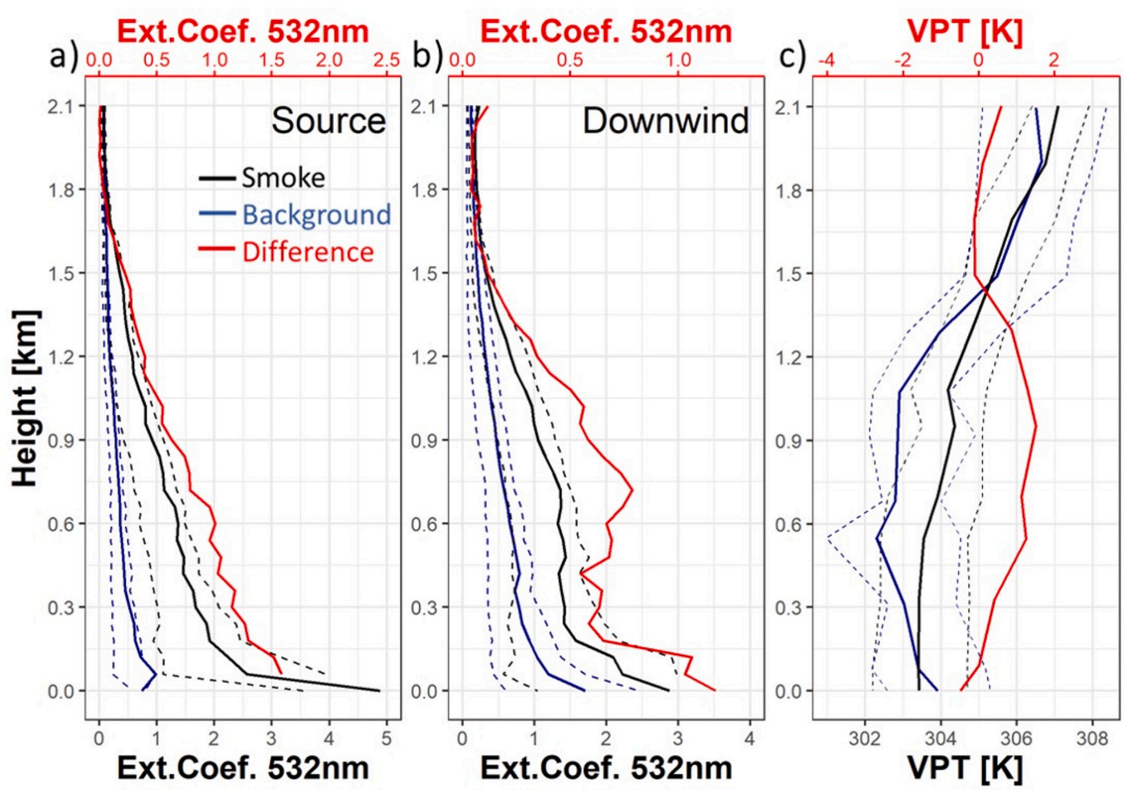

Fig. 3. Vertical distribution of aerosols (extinction coefficient at 532 nm) over the source (a) and downwind (b) regions during smoky days (solid black line) and background (solid blue line). The dashed line represents the 25th and 75th percentile for background (blue) and smoke-dominated (black) days, while the solid red lines represent the smoke and background extinction coefficient differences plotted on the secondary x-axis. (c): The vertical profiling of atmospheric virtual potential temperature (VPT) was obtained from the radiosonde station in Delhi during smoke (black line), background days (blue line), and the difference on the secondary x-axis (red line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

vertical profile of aerosols showed a gradual decrease in extinction coefficient values with height during the background period, which is characteristic of surface anthropogenic emissions over this region. Similarly, (Jethva et al., 2018a) also reported a higher near-surface CALIPSO-based backscatter coefficient decreasing with altitude over both Delhi and Punjab, with a relatively higher magnitude over Delhi during the post-monsoon season.

On the other hand, the near-surface (below 250 m) extinction coefficient value over the source region was ~6 times higher during CRB-smoky days, with the surface value reaching close to 5 km^{-1} at 532 nm and around 7 km^{-1} at 1064 nm. The dense smoke emissions from biomass burning significantly increase the near-surface light extinction, which is also in agreement with CRB-induced enhancement in $PM_{2.5}$ concentrations over source region (Fig. 2a). Although an enhancement in near-surface aerosol extinction coefficient was clearly seen over the downwind region during CRB days (compared to the background values), though in smaller magnitude. This is in line with our anticipation of decreasing smoke concentration with distance as we move farther from the source. However, such similar pattern was not observed in the enhancement of $PM_{2.5}$ in Fig. 2. As explained below, $PM_{2.5}$ enhancement increases with distance from the source, suggesting that direct transport of $PM_{2.5}$ associated with CRB alone cannot explain the enhanced $PM_{2.5}$ concentration over the downwind region.

Although the extinction coefficient of aerosols gradually decreased with height during CRB days over the source region, the differences in the magnitude of the extinction coefficient values indicate a distinct enhancement of 100–200% from 250 m till 1 km altitude. In comparison, an interesting scenario emerges over the downwind region; a distinct elevated aerosol layer is present from 250 to 1000 m altitude above the surface with an enhancement of 200–300% (compared to background) and a relatively high extinction coefficient ranging (1–1.5 km^{-1}) at 532 nm (0.6–0.75 km^{-1} at 1064 nm). The CRB-induced enhancement of elevated layers over downwind region suggests that the smoke plumes are rising higher in altitude as they travel downwind from the source emission region.

Further, we analyzed the Indian Meteorological Department (IMD) radiosonde profiles launched at 1730 LT every day over Delhi from 2007 to 2020. Corresponding to the classification used in Fig. 2, these radiosonde observations during the CRB season are segregated into CRB-smoky days and background days and compared (Fig. 3c). The magnitude of mean virtual potential temperature (VPT) decreases with height during background days from surface till 800 m, which suggests unstable atmospheric conditions and strong turbulent mixing conditions. Above 800 m till 1200 m, the VPT values remain constant with height, indicating neutral conditions. Above 1200 m, the atmosphere becomes stable and indicates the transition from a mixing layer to a free troposphere. A simple application of the parcel method for mixing height determination (Seibert, 2000) also indicates that any parcel released from the surface can go up to ~1200 m under background conditions, suggesting a mean mixing layer of ~1.2 km. In contrast, the magnitude of mean VPT increased with the height from the surface up to 800 m altitude during smoky days over Delhi, which suggests inversion conditions and a stable atmosphere. A simple application of the parcel method for mixing height determination indicates that any parcel released at the surface during smoky days will not rise, and mixing layers will be very shallow.

Next, the observational relationship between co-temporal mixing layer height, surface $PM_{2.5}$ concentrations, and columnar aerosol loading (AOD) over Delhi is analyzed (Fig. 4a-b). Daily mixing layer heights were estimated from the radiosondes using the parcel method (details in method section). The hourly $PM_{2.5}$ concentration at the US consulate at New Delhi is averaged over the span of time 10:00 to 14:00 UTC to match the radiosonde measurements at 12:00 UTC. As expected, the $PM_{2.5}$ concentration is proportional to a decrease in the mixing height ($R = -0.82$) during the CRB period, i.e., more particulate matter is accumulated near the surface at a lower mixing height. Similarly, a robust negative correlation ($R = -0.74$) is observed between MAIAC AOD and mixing height (Fig. 4b). These results (Figs. 3 and 4) indicate that the elevated smoke layers over downwind region during the CRB period may induce atmospheric stability and suppress the boundary layer turbulence, which will subsequently lead to an extremely polluted scenario as the surface emissions start accumulating within the surface layer.

However, this certainly gives the subject thought to whether these elevated smoke layers are so impactful. The reanalysis product, MERRA-2, incorporates anthropogenic and CRB emissions and assimilates AOD observations to provide a simulated 3-D dataset of the aerosol concentrations and their radiative effect on meteorology over our study region.

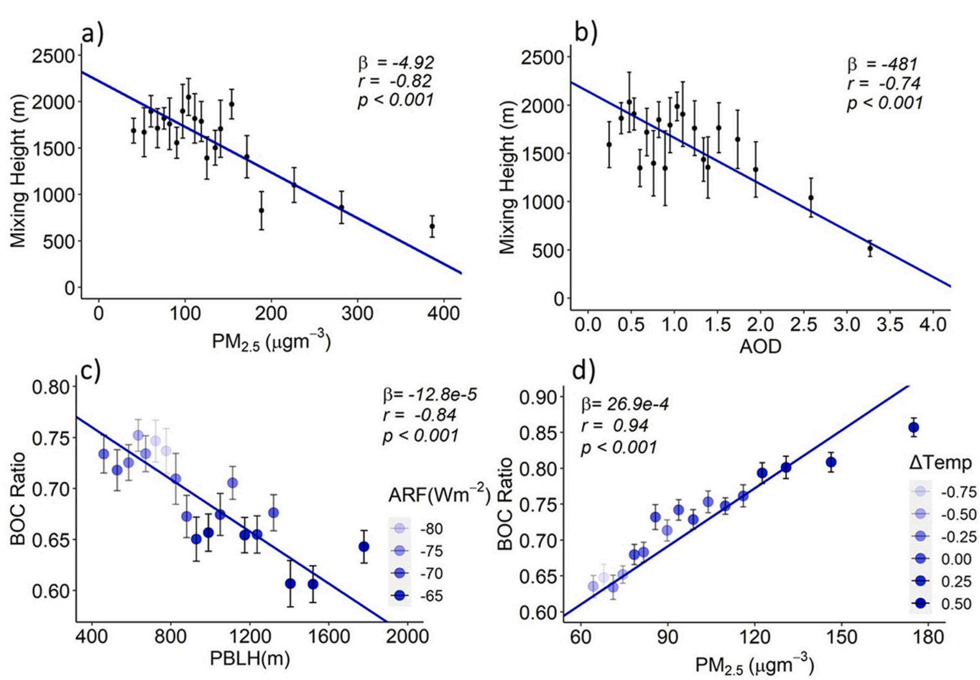

Fig. 4. Scatterplot of radiosonde derived mixing layer height versus a) $PM_{2.5}$ concentrations binned into 20 bins each 5% percentile of $PM_{2.5}$, b) AOD binned into 20 bins 5% percentile each. β, r & *p* represent the slope of the reduced major axis (RMA) regression line, correlation coefficient, and the *p*-value respectively. (c&d) show scatterplots of MERRA-2 black and organic carbon ratio at 940 hPa to near-surface versus c) MERRA-2 PBLH binned into 20 bins 5% percentile each, d) MERRA-2 $PM_{2.5}$ concentrations binned into 20 bins each 5% percentile each. ΔTemp represents the mean of air temperature differences between ~940hpa and ~ 985hpa for each $PM_{2.5}$ bin and the ARF represent the mean ARF for each PBLH bin. The blue solid lines in all give the reduced major axis (RMA) regression for each two datasets. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Here, MERRA-2 simulated BOC ratio during the CRB season is segregated according to the composites of CRB-smoky days and background days used in Fig. 2 (Fig. S6). During smoky days, MERRA-2 showed a higher BOC ratio (0.8 ± 0.1) than background days (0.6 ± 0.1), indicating that MERRA-2 simulations were able to capture the CRB smoke transport and associated elevated smoke over Delhi region. In response to the increasing carbonaceous aerosols in the atmospheric column, the shortwave heating rate was significantly higher during smoky days than background days (Fig. 5). The heating rate profiles showed a much higher shortwave heating rate during smoky days with peaks at ~900hpa-850hpa while the near-surface heating rate was higher during background days. The downwind region showed a relatively higher shortwave heating rate at a higher altitude (875hpa-850hpa) compared to source region. The vertical stratification of the difference between smoke and background heating rate was prominent over downwind region with a higher difference at higher altitude (850hpa) than near-surface. This suggests that absorbing shortwave solar radiation at higher altitudes relatively more over the downwind region induces a strong dimming effect at the surface and reduces the near-surface heating rate.

Thus, MERRA-2 dataset was further used to examine and understand the simulated response of midday means boundary layer height, surface $PM_{2.5}$ concentrations, direct aerosol radiative forcing (ARF), and atmospheric stability (Temp940hpa-Temp985hpa) to increase in BOC ratio is examined (Fig. 4 c&d). A significant negative correlation was found between the BOC ratio and mixing layer height corresponding to higher negative ARF at a higher BOC ratio (Fig. 4c). Increasing the carbonaceous aerosols at higher altitudes (high BOC ratio) enhanced the upper-level absorbing of solar radiation (higher heating rate Fig. S6) and induced surface dimming, weakening the surface heat fluxes. This leads to suppression of the mixing layer and derives a stronger boundary layer cooling effect (negative ARF). On the other hand, $PM_{2.5}$ showed a significantly positive correlation in response to increasing BOC ratio and increasing the upper layer temperature (positive ΔTemp) (Fig. 4d). Finally, the elevated smoke layer enhances the atmospheric stability and induces inversion conditions leading to accumulation of the pollutants near-surface. In agreement, higher $PM_{2.5}$ concentration (~ 135 μg/m^3) and shallower PBLH (~300 m) was simulated by MERRA2 during smoky days (Fig. S6). Also, the ARF values (negative) was estimated as 100 ± 24 Wm^{-2} during the smoky days in comparison to 64 ± 16 Wm^{-2} during background days suggesting a higher cooling effect with the presence of elevated smoke layer (see Fig. S6b).

4. Discussion

This study examines the direct and semi-direct effects of the smoke plume from crop residue burning (CRB) fires on the $PM_{2.5}$ variability over the CRB source and downwind regions. Long-term data from multi-satellite observations (2007–2020), ground-based in-situ observations (2015–2020), and IMD radiosonde profiles (2007–2020) are synergistically used to improve our understanding of the impact of CRB aerosols on the air quality over Megacity Delhi. First, the most probable region through which CRB emitted smoke plumes travel from the source region to Delhi (downwind region) was identified based on the spatial pattern

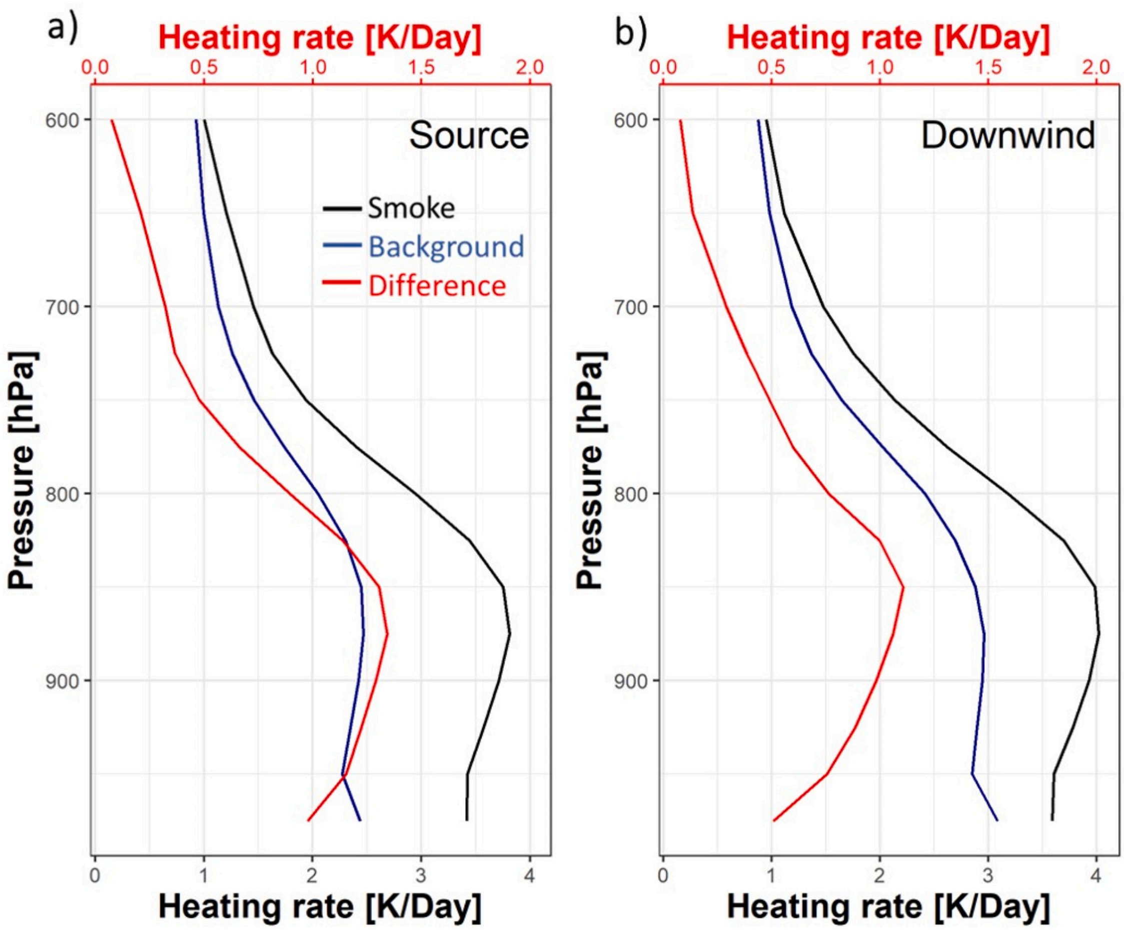

Fig. 5. Shortwave heating rate profile (Temperature tendency, in K Day-1) over the source (a) and downwind (b) regions during smoky days (solid black line) and background (solid blue line). The solid red lines represent the smoke and background extinction coefficient differences plotted on the secondary x-axis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of extreme AOD, AAOD, and UV-AI values within the region (CRB_{box}). Next, we used the AOD variability over the CRB_{box} region to classify all our data into smoky and background (non-smoky) days, i.e. the days with very high AOD (> 75 percentile) over CRB_{box} are considered as CRB-smoke affected days; other days are indexed as background days when CRB_{box} averaged AOD < 25 percentile. Further, differences in surface $PM_{2.5}$, aerosol vertical distribution, atmospheric stability, and mixing height between the smoky and background days were analyzed to gain a mechanistic understanding of the aerosol-meteorology-$PM_{2.5}$ coupling over the downwind receptor region.

A distinct enhancement in daily mean $PM_{2.5}$ values was observed over monitoring sites inside the CRB_{box} (~100–200 μgm^{-3} or ~ 150–200%) during smoky days, while the same was ~50% (20–75 μgm^{-3}) outside the CRB_{box}. Within CRB_{box}, strong spatial heterogeneity was observed, with the magnitude of $PM_{2.5}$ enhancement over downwind region showing relatively larger values. Detailed investigation revealed that the smoke-associated enhancement in $PM_{2.5}$ over the downwind region increases linearly from midnight and peak around noon hours. At the same time, the investigation of the vertical distribution of aerosols illustrated a distinct and robust elevated aerosol layer located at altitude ~300–800 m (with high Ext. Coef. = 1–1.5 km^{-1}) over downwind region during smoky conditions (but gradually decrease over source region). The analysis indicates that the portion of smoke of plumes injected to a higher altitude and move downwind from the emission region (towards Delhi), mainly due to self-lofting (Jethva et al., 2018b; Gonzalez-Alonso et al., 2019; Liu et al., 2020). Simultaneous analysis of radiosonde profiles indicates that these elevated smoke layers over Delhi can induce strong inversion conditions (below 1 km altitude) and reduce the mixing height. Thus, all the local anthropogenic emissions over Delhi are trapped over the city and eventually result in dense $PM_{2.5}$ or haze episodes during the CRB season. Thus, in addition to the direct advection of particles and gases from CRB region, the semi-direct impacts of these smoky elevated aerosol layers can also contribute to substantial enhancement in $PM_{2.5}$ particles over Delhi. Additionally, the seasonal shift in the CRB activities from Oct to Nov (early winter) after implementing the groundwater preservation policy in 2009 also can strengthen the temperature inversion caused by the elevated heating layer due to smoke aerosol absorption leading to the trapping of air pollutants near-surface. Further, this elevated aerosol-induced stagnation can trigger other indirect contributions to the particulate matter enhancement like the secondary aerosol formation through the enhanced gas-to-particles conversion process of accumulated precursor gases emitted within Delhi (Guo et al., 2014; Kulmala et al., 2020). The new particle formation can even be the dominant contributor to haze formation in urban areas and account for as high as 50% of aerosol mass concentration (Kulmala et al., 2020; Casquero-Vera et al., 2021; Lalchandani et al., 2021; Reyes-Villegas et al., 2021).

Nevertheless, our methodology and analysis have some inherent uncertainties. First, satellite retrievals of aerosol properties (AOD and AAOD), and active fire are subjected to missing data due to cloud cover, very heavy smoke (Mhawish et al., 2019) and undetected fire (for fire detection). Note that the very small size agriculture fires (<0.01% of the MODIS pixel area) also can't be detected in MODIS 1km^2 footprint (Wooster et al., 2005; Giglio et al., 2016). As such, the CRB smoke box daily mean MAIAC AOD values were used as proxy to CRB fire effect. Note that MAIAC has statistically significant association with CRB active fire count within the smoke box at daily scale (Fig. S2). In addition, MAIAC showed a good ability to retrieve AOD on a small scale between the cloud and thick smoke layers (Mhawish et al., 2019). Moreover, the radiosonde derived mixing layer estimates using Parcel method may be underestimated (Seidel et al., 2010), but biases in magnitudes will not impact our process-level analysis. Secondly, the long-term ground-level $PM_{2.5}$ monitoring stations are unevenly distributed over this region, with fewer sites in the CRB source region. Similarly, the numbers of samples of the vertical aerosol profiles and corresponding radiosondes are also limited due to the coarse temporal resolution of the CALIPSO passes over our study region. These sampling limitations may affect the quantitative values reported in our analysis over the source and downwind regions.

To sum up, this is the first observational analysis that shows robust observational AOD-$PM_{2.5}$-PBLH coupling, via presence of elevated aerosol layers over Delhi during the extreme haze episodes. Although, regulating the CRB activities at source is the solution of this air quality problem, CRB being a socio-economic challenge, the inherent human-induced randomness and widespread area of the agricultural land limits the efficiency of sudden policy-based control of the CRB emissions. In this context, our findings indicate that administrating special curtailment/lockdowns of the local emissions over Delhi during CRB season (i.e. during mid-Oct to mid-Nov) can also be strategically successful for mitigating occurrences of hazardous $PM_{2.5}$ values during extreme CRB emissions over Delhi. Overall, presence of these smoky elevated layers has strong implications on regional air quality modeling/operational forecasting during extreme $PM_{2.5,}$ haze and fog events (Kumar et al., 2020; Jena et al., 2021) and warrants more detailed observational campaigns in future.

Credit author statment

CS conceptualized the study. AM and CS performed the data analysis. AM led the figure manuscript writing and revision process with regular inputs from all other co-authors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2022.113167.

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